

## Elevator Pitch

E-Commerce Sales Dashboard - an interactive Power BI dashboard built for a US-based e-commerce company to monitor Year-to-Date sales, profit, units sold and profit margin across segments, categories and states. Data is stored in MS SQL Server (imported from Kaggle CSVs), cleaned in Power Query, modeled with a Data table and relationships, and visualised with dynamic KPIs that change colour and show trend icons based on YoY performance. The Dashboard helps stakeholders spot winning products, underperforming regions, and shipping methods efficiency at a glance

## Step by step walkthrough project

1. **Repos & assets** : GitHub repo , dataset, visuals and documentation are available
2. **Problem & Objective** :
  1. Business need : quickly measure YTD performance and identify growth/ decline across customers (Consumer/Corporate/Home Office), categories, states and shipping methods
  2. Outcome : enable data-driven promotion decisions, inventory & shipping strategy
3. **Data Ingestion** :
  1. Source : CSV files(kaggle dummy dataset) -> imported into MS SQL Server (Created Ecommerce\_Database and tables).
  2. Why SQL : central storage, easier to refresh and scale vs raw CSV
4. **Power BI connection** :
  1. Connected Power BI desktop to the MS SQL database and loaded the e-commerce table + state lat/long table.
5. **Power Query** - data cleaning :
  1. Converted column types (dates, decimals, integers, text)
  2. Trimmed whitespace, renamed columns, removed unnecessary columns, replaced nulls or filled defaults , and removed duplicates.
  3. Created / standardised keys (eg. customer\_state) to join with geo table.
6. **Data Modelling** :
  1. Created a dedicated Calendar (Date) table.
  2. Relationships :
    1. Calendar[Date] (1) -> ecommerce\_data[order\_date] (many)
    2. us\_state\_long\_lat\_codes[name] (1) -> ecommerce\_data[customer\_state] (many)
  3. Why : enables TOTALYTD, SAMEPERIODLASTYEAR, slicer syncs and accurate filtering.
7. **DAX & KPIs (core logic)**
  1. Created measures : YTD Sales, PYTD Sales, YoY Sales Growth, YTD Profit, YTD Quantity, Profit Margin, etc
  2. Built dynamic icons with UNICAR() and dynamic color measures for conditional formatting.
  3. Eg : uses TOTALYTD, CALCULATE, DATESYTD, SAMEPERIODLASTYEAR.
8. **Visuals & interaction**
  1. KPI banner (top) : 4 cards with YTD Sales, Profit, Quantity, Profit Margin. Each card shows mini trend and changes background colour to green/red depending on YoY direction.
  2. Slicer : customer segment (Consumer/Corporate/Home office) at top-right (tiled). Slicer drives whole page.
  3. Sales by Category (matrix) with YTD / PYTD / YoY and arrow icons.
  4. Sales by State : map/ bubble (latitude / longitude) where bubble size = YTD Sale ; colour by region
  5. Top 5 / Bottom 5 products: bar charts.
  6. Donut: Sales by Region, and Shipping type analysis.

## 9. Insights & Value

1. Quick decisions: which regions/products to promote or discount, shipping method optimization, inventory focus.
2. Example insight: "Top 5 products contribute X% of sales; West region has highest profit margin; standard shipping yields highest customer satisfaction/cost tradeoff."

## 10. Deliverable

1. Power Bi .pbix (report), the SQL import scripts or instructions, and the README & PDF for handover.

## Some Definitions

1. **KPI (Key Performance Indicator)** : A Measurable value that indicates how effectively a business is achieving key objectives (e.g., YTD Sales)
2. **KPI Banner** : Top-row collection of KPI cards providing an at-a-glance summary (YTD Sales, Profit, Qty, Margin)
3. **Power BI** : Microsoft editor inside Power BI used for cleaning and transforming raw data before loading to the model.
4. **Power Query** : ETL editor inside Power BI used for cleaning and transforming raw data before loading to the model
5. **MS SQL Server** : Relational database to store and query structured data; useful for centralising datasets and scheduled refreshes
6. **Data Modelling** : Defining tables & relationships (onr-to-many, many-to-one) so slicers/aggregations behave correctly
7. **Calendar / Date Table** : A separate table with continuous dates used as the authoritative date for time intelligence.
8. **DAX ( Data Analysis Expressions )** : Power BI formula language used for creating measures and calculated columns (e.g., TOTALYTD).
9. **TOTALYTD / SAMEPERIODLASTYEAR**: Time-intelligence functions to compute year-to-date numbers and compare with previous year.
10. **UNICHAR()** : Function to show special characters (used for up/down arrow icons)
11. **Conditional Formatting** : Dynamically changing color or background based on measure (green for increase, red for decrease).
12. **Slicers / Cross-Filtering**: Visual controls to filter page; interactions can be set to highlight or filter visuals

## Questions

Q: What was the biggest data quality issue and how did you fix it?

A: "State names had inconsistent casing/extra spaces which broke joins with the geo table. I used Power Query to Trim() and Proper() the fields and de-duplicated records; also standardized data types to avoid type errors."

Q: Why a Date table?

A: "Date table is best practice for accurate time intelligence. It provides continuous date range, fiscal calendar columns if needed, and enables TOTALYTD/SAMEPERIODLASTYEAR functions to work correctly."

Q: Why use MS SQL Server instead of loading CSV into Power BI?

A: "SQL Server centralizes data, enables scheduled refreshes, supports larger datasets, and allows pre-processing via SQL if needed. It also works better for multi-user environments."

Q: How does conditional formatting work in your KPI?

A: "I created a color/indicator measure (returns 'Green'/'Red' or UNICHAR arrows) and used it in the card/shape conditional formatting or field formatting for background color and icon."

Q: What would you improve if you had more time?

A: "Automate ETL with scheduled jobs, add row-level security for segment-based access, implement incremental refresh for large datasets, and add anomaly detection (outlier alerts) for sales drops."

Q: How do you ensure your model is performant?

A: "Use star schema (fact + dimension tables), avoid calculated columns on huge tables where possible, use measures instead of row calculations, and enable query reduction options and aggregations if needed."

## Challenges

- Fixed mismatched join keys between state table and orders (cleaned in Power Query).
- Built efficient DAX using VAR to prevent repeated computation.
- Created intuitive conditional formatting for business readability.